

Troubleshooting Access Issues

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Systematic Diagnosis of IAM Problems

Access issues are among the most common support requests. This notebook provides a systematic methodology for diagnosing and resolving IAM-related problems including permission denials, policy conflicts, and boundary issues.

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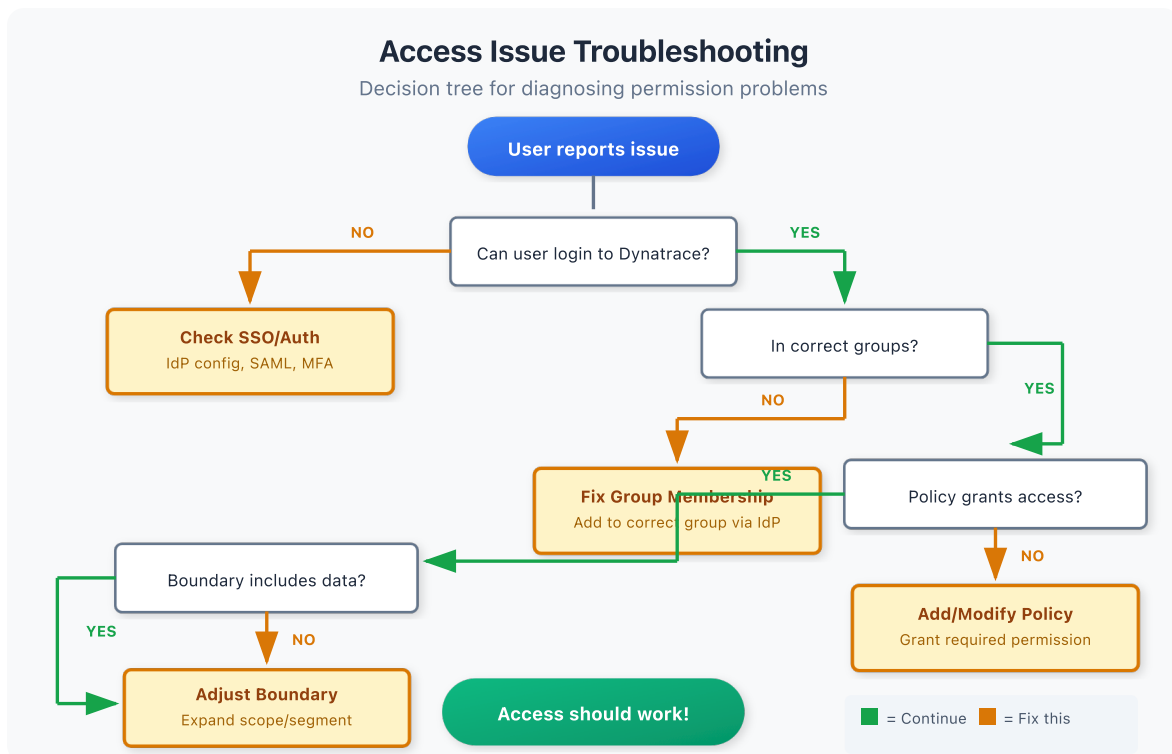
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Prerequisites

Requirement	Details
Dynatrace Environment	SaaS with Gen3 IAM
Permissions	<code>account-iam-admin</code> for full diagnosis
Knowledge	IAMADM-01 through 08 completed

1. Troubleshooting Methodology

Follow a systematic approach to diagnose access issues.



Step-by-Step Methodology

Step	Check	Tools
1	User can authenticate	SSO logs, IdP logs
2	User exists in Dynatrace	Account Management → Users
3	User is in correct groups	Account Management → Groups
4	Groups have correct policies	Account Management → Policies
5	Policies grant required permissions	Policy viewer
6	Boundaries don't restrict access	Boundary configuration
7	No conflicting policies	Policy evaluation
8	Data exists and is accessible	DQL query

Information Gathering

Before troubleshooting, collect:

User Information:

- ☐ User's email address
- ☐ What action they're trying to perform
- ☐ Exact error message received
- ☐ When the issue started
- ☐ What changed recently (role, team, etc.)

Environment Information:

- ☐ Which environment(s) affected
- ☐ What data/resource they're accessing
- ☐ Screenshot of error (if available)

2. Common Access Issues

Issue Categories

Category	Symptoms	Common Cause
Authentication	Can't login	SSO misconfigured, account disabled
Authorization	Access denied	Missing policy, wrong group
Data Access	No results	Boundary restriction, no data
Partial Access	See some, not all	Segment restrictions
Intermittent	Sometimes works	Caching, eventual consistency

Quick Diagnosis Table

Symptom	Likely Cause	First Check
"Access Denied"	Missing permission	Policy assignments
"Forbidden"	Boundary restriction	Boundary config
"Not Found"	No data or no access	Query without filters
"Unauthorized"	Token/auth issue	Token validity
Can't see entity	Segment restriction	Segment coverage
Can read, can't write	Read-only policy	Policy permissions

3. Permission Denied Debugging

When users receive "Access Denied" or "Forbidden" errors.

Diagnostic Steps

Step 1: Verify User Exists

1. Go to Account Management → Users

2. Search for user by email
3. Verify account is active (not disabled)

Step 2: Check Group Membership

1. Click on user to view details
2. Note all groups user belongs to
3. Verify expected groups are present

Step 3: Review Group Policies

1. For each group, check assigned policies
2. Verify policies grant needed permissions
3. Check policy conditions/restrictions

Step 4: Check Boundaries

1. Review boundaries assigned to groups
2. Verify boundaries include target resources
3. Check boundary conditions

Permission Evaluation Order

1. User authenticated? → No = Auth failure
2. User in any groups? → No = No permissions
3. Any policy grants permission? → No = Access denied
4. Any boundary restricts? → Yes = Check boundary scope
5. Access granted

Common Permission Issues

Issue	Symptom	Resolution
Not in group	No access at all	Add user to group via IdP
Wrong group	Partial access	Move to correct group
Policy too restrictive	Can't perform action	Modify or add policy
Missing environment	Can't access env	Add env to policy scope

4. Policy Conflict Resolution

When multiple policies interact unexpectedly.

How Policy Evaluation Works

1. Collect all policies from all user's groups
2. Evaluate each policy's conditions
3. **Any ALLOW** = access granted (unless explicitly denied)
4. **No matching ALLOW** = access denied

Conflict Scenarios

Scenario	Policy A	Policy B	Result
Both allow	ALLOW read	ALLOW write	ALLOW both
One allows	ALLOW read	No match	ALLOW read
Overlapping	ALLOW all envs	ALLOW prod only	ALLOW all envs
Condition mismatch	ALLOW (condition: A)	ALLOW (condition: B)	Depends on conditions

Debugging Policy Conflicts

Step 1: List All Applicable Policies

1. Identify all groups user belongs to
2. List all policies assigned to those groups
3. Note overlapping permissions

Step 2: Evaluate Each Policy

For each policy, check:

- Does it apply to the target resource?
- Does it grant the required permission?
- Are conditions met?

Step 3: Identify Conflict

Conflict Type	Indicator	Resolution
Unintended restriction	User in wrong group	Remove from group
Missing permission	No policy grants access	Add policy
Condition not met	Policy exists but doesn't apply	Adjust condition
Scope mismatch	Policy applies to wrong scope	Correct scope

5. Boundary Troubleshooting

Boundaries restrict what data users can access within their permissions.

Boundary Evaluation

```

User wants to access data
↓
Policy grants permission? → No = Access denied
↓ Yes
User has boundary assigned? → No = Access all data
↓ Yes
Data within boundary scope? → No = Data hidden
↓ Yes
Data returned

```

Common Boundary Issues

Issue	Symptom	Cause	Resolution
Too restrictive	See no data	Boundary too narrow	Expand boundary
Not restrictive enough	See too much	Boundary too broad	Narrow boundary
Segment mismatch	Some data missing	Segment doesn't match	Fix segment definition
Environment excluded	No env access	Env not in boundary	Add environment

Boundary Debugging Steps

Step 1: Identify Boundary

1. Find user's groups
2. Check boundaries assigned to groups
3. Note boundary scope (environment, storage, settings)

Step 2: Verify Boundary Scope

For each domain:

- **Environment:** Which environments included?
- **Storage:** Which data types included?
- **Settings:** Which settings accessible?

Step 3: Test Without Boundary

Temporarily:

1. Add user to unbounded admin group
2. Test if access works
3. If yes, boundary is the issue
4. Remove from admin group

Step 4: Adjust Boundary

Modify boundary to include:

- Missing environments
- Missing segments
- Missing data types

6. SSO and Authentication Issues

When users can't login at all.

SSO Troubleshooting

Symptom	Check	Resolution
Redirect loop	IdP configuration	Verify ACS URL
SAML error	SAML response	Check attribute mapping
User not found	JIT provisioning	Enable auto-provisioning
MFA failure	MFA configuration	Verify MFA setup
Session expired	Session settings	Extend session duration

SAML Debugging

Check SAML Response:

1. Use browser dev tools → Network
2. Find SAML response during login
3. Decode and verify:
 - Subject/NameID (email)
 - Groups attribute
 - Conditions (timestamps)

Common SAML Issues:

Issue	Indicator	Fix
Wrong NameID	Email doesn't match	Configure NameID format
Missing groups	Groups not in assertion	Add groups attribute in IdP
Expired assertion	Time mismatch	Sync clocks, extend validity
Invalid signature	Signature error	Update IdP certificate

Group Mapping Issues

If user logs in but has wrong permissions:

1. Check SAML groups in assertion
2. Verify group mapping configuration

3. Confirm IdP group names match exactly
4. Check for case sensitivity

7. Token and API Access Issues

When API tokens or OAuth clients don't work as expected.

Token Issues

Error	Cause	Resolution
401 Unauthorized	Token invalid/expired	Regenerate token
403 Forbidden	Missing scope	Add required scope
404 Not Found	Wrong endpoint or no data	Verify API path
Token rejected	Token revoked	Create new token

Token Debugging Steps

Step 1: Verify Token Valid

1. Check token hasn't expired
2. Verify token not revoked
3. Confirm token is for correct environment

Step 2: Check Scopes

1. List token's scopes
2. Compare to required scopes for API
3. Add missing scopes if needed

Step 3: Test with curl

```
curl -H "Authorization: Api-Token <TOKEN>" \
https://<env>.live.dynatrace.com/api/v2/entities
```

OAuth Client Issues

Issue	Check	Resolution
Can't get token	Client credentials	Verify client ID/secret
Wrong permissions	Client scopes	Add required scopes
Token expired	Token lifetime	Request new token

8. Diagnostic Queries

DQL queries to help diagnose IAM issues.

```
// Find recent access denied events for specific user
fetch logs, from: now() - 24h
| filter matchesPhrase(log.source, "audit")
| filter matchesPhrase(content, "denied") or matchesPhrase(content,
"forbidden")
| fields timestamp, content
| sort timestamp desc
| limit 50
```

```
// User's recent authentication events
fetch logs, from: now() - 7d
| filter matchesPhrase(log.source, "audit")
| filter matchesPhrase(content, "login") or matchesPhrase(content,
"authentication")
| fields timestamp, content
| sort timestamp desc
| limit 100
```

```
// Failed authentication attempts
fetch logs, from: now() - 7d
| filter matchesPhrase(log.source, "audit")
| filter matchesPhrase(content, "failed") and (matchesPhrase(content,
"login") or matchesPhrase(content, "authentication"))
| fields timestamp, content
| sort timestamp desc
| limit 50
```

```
// Recent policy or group changes that might affect access
fetch logs, from: now() - 7d
| filter matchesPhrase(log.source, "audit")
| filter (matchesPhrase(content, "policy") or matchesPhrase(content, "group")
or matchesPhrase(content, "boundary"))
| filter matchesPhrase(content, "modified") or matchesPhrase(content,
"created") or matchesPhrase(content, "deleted")
| fields timestamp, content
| sort timestamp desc
| limit 50
```

```
// Token-related events
fetch logs, from: now() - 7d
| filter matchesPhrase(log.source, "audit")
| filter matchesPhrase(content, "token")
| fields timestamp, content
| sort timestamp desc
| limit 50
```

```
// All audit events in last hour (for immediate troubleshooting)
fetch logs, from: now() - 1h
| filter matchesPhrase(log.source, "audit")
| fields timestamp, content
| sort timestamp desc
| limit 200
```

Troubleshooting Checklist

Use this checklist when debugging access issues:

AUTHENTICATION

- ☐ User can access IdP login page
- ☐ User can authenticate with IdP
- ☐ SAML assertion is valid
- ☐ User account exists in Dynatrace
- ☐ User account is active (not disabled)

GROUP MEMBERSHIP

- ☐ User is in expected groups
- ☐ Groups are synced from IdP (if using SCIM/JIT)
- ☐ No unexpected group memberships

POLICIES

- ☐ Groups have policies assigned
- ☐ Policies grant required permissions
- ☐ Policy conditions are met
- ☐ No conflicting policies

BOUNDARIES

- ☐ Boundary includes target environment
- ☐ Boundary includes target data type
- ☐ Segment definitions are correct
- ☐ No overly restrictive boundaries

DATA

- ☐ Data exists in target timeframe
- ☐ Data matches segment criteria
- ☐ Query syntax is correct

Escalation Path

When self-service troubleshooting doesn't resolve the issue:

Level	Who	When
L1	Team lead	Can't determine cause
L2	IAM admin	Need policy/boundary changes
L3	Platform team	Infrastructure/IdP issues
L4	Dynatrace support	Product issues

Information for Escalation

Include in escalation:

- ☐ User email/identifier
- ☐ Exact error message
- ☐ Steps to reproduce
- ☐ Timestamp of issue
- ☐ What was already checked
- ☐ Audit log excerpts
- ☐ Screenshots

Series Conclusion

Congratulations! You've completed the IAMADM series.

Series Recap

Notebook	Key Topics
01: Governance Foundations	Framework, roles, models
02: SSO and Authentication	SAML, IdP integration, MFA
03: Group Architecture	Hierarchy, naming, SAML mapping
04: Policy Authoring	Syntax, conditions, GitOps
05: Boundary Design	Three domains, isolation patterns
06: User Lifecycle	SCIM, JIT, tokens
07: Audit and Compliance	Audit queries, SOC2/SOX/HIPAA
08: Multi-Environment	Account vs env, break-glass
09: Troubleshooting	Methodology, common issues

Ongoing Maintenance

Task	Frequency
Access reviews	Quarterly
Policy review	Semi-annually
Boundary review	Semi-annually
Token rotation	Per policy (90 days typical)
Audit log review	Monthly
Documentation update	As needed

Summary

In this notebook, you learned:

- Systematic troubleshooting methodology
- Common access issue categories
- Permission denied debugging steps
- Policy conflict identification and resolution
- Boundary troubleshooting techniques
- SSO and authentication debugging
- Token and API access troubleshooting
- Diagnostic DQL queries

References

- [IAM Troubleshooting](#)
 - [SAML Configuration](#)
 - [API Authentication](#)
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